
SSM Optimization

Weekly Meeting

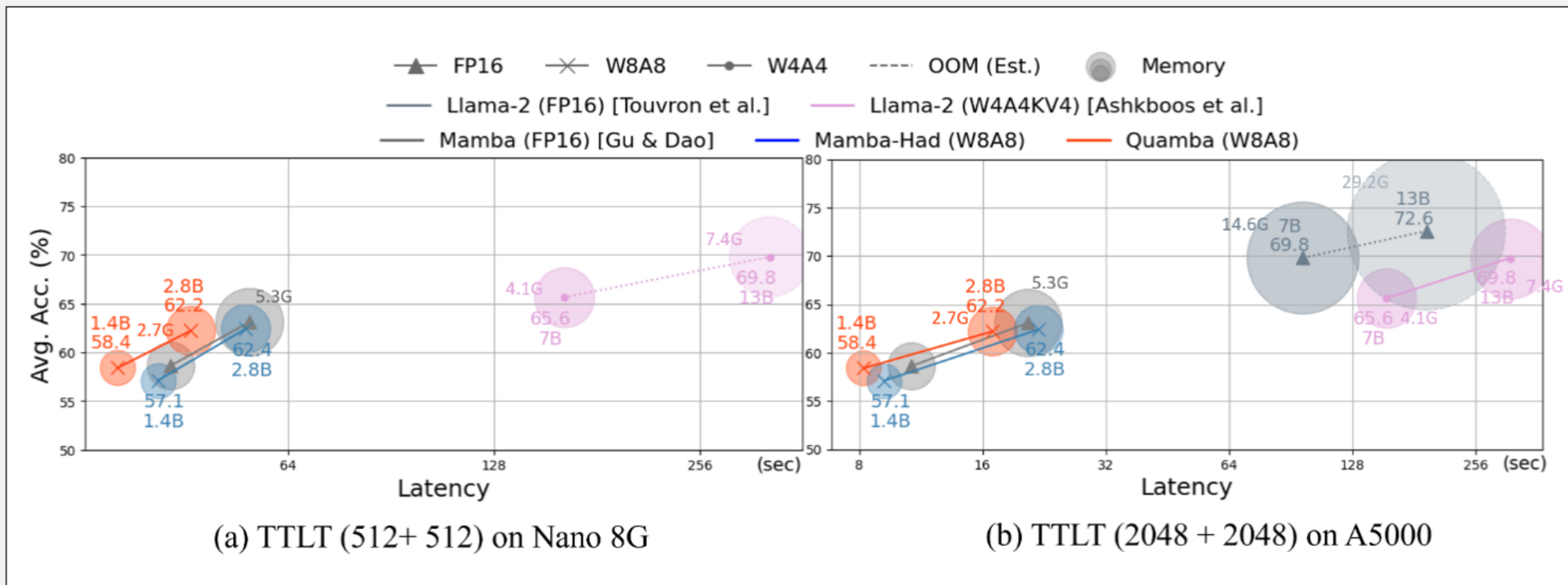
20/02/26

Quamba

ICLR 2025

- Per-tensor 8-bit quantization for SSMs
- Key aspects:
 1. Percentile-based ($p=99.999$) **max values suppression of the input** activations (few small outliers, $< 0.1\%$)
 2. Quantize the **output activations** in an outlier-free space with **Hadamard transform** (outliers appear in the SSM output)
- Calibration set: from Pile dataset
- Target architectures: Mamba 2.8B, Jamba 52B (self-attention + SSMs + MoE)
- Baselines: static, dynamic quantization and Mamba-PTQ + SmoothQuant, QuaRot on A5000 (24GB) and Orin Nano 8GB

Quamba



Quamba2

ICML 2025

- Key aspects:
 - They observe consistency in the activated channels and states across time steps and input samples.
 - 1. Cluster-aware **weight reordering** to group SSM heads and channels with similar value ranges and to scaling factors per blocks (requires calibration)
 - 2. State persistence for input-dependent parameters (B,C) → quantization applied **per-state group**
- They also consider mixed-precision quantization (via evolutionary search) for the activations (W4A{8/16}) with a fixed A8:A16 ratio of 3:1

Next steps

- KL divergence analysis
- Test RTN
- Evaluate on hybrid architectures
- Check baselines' code